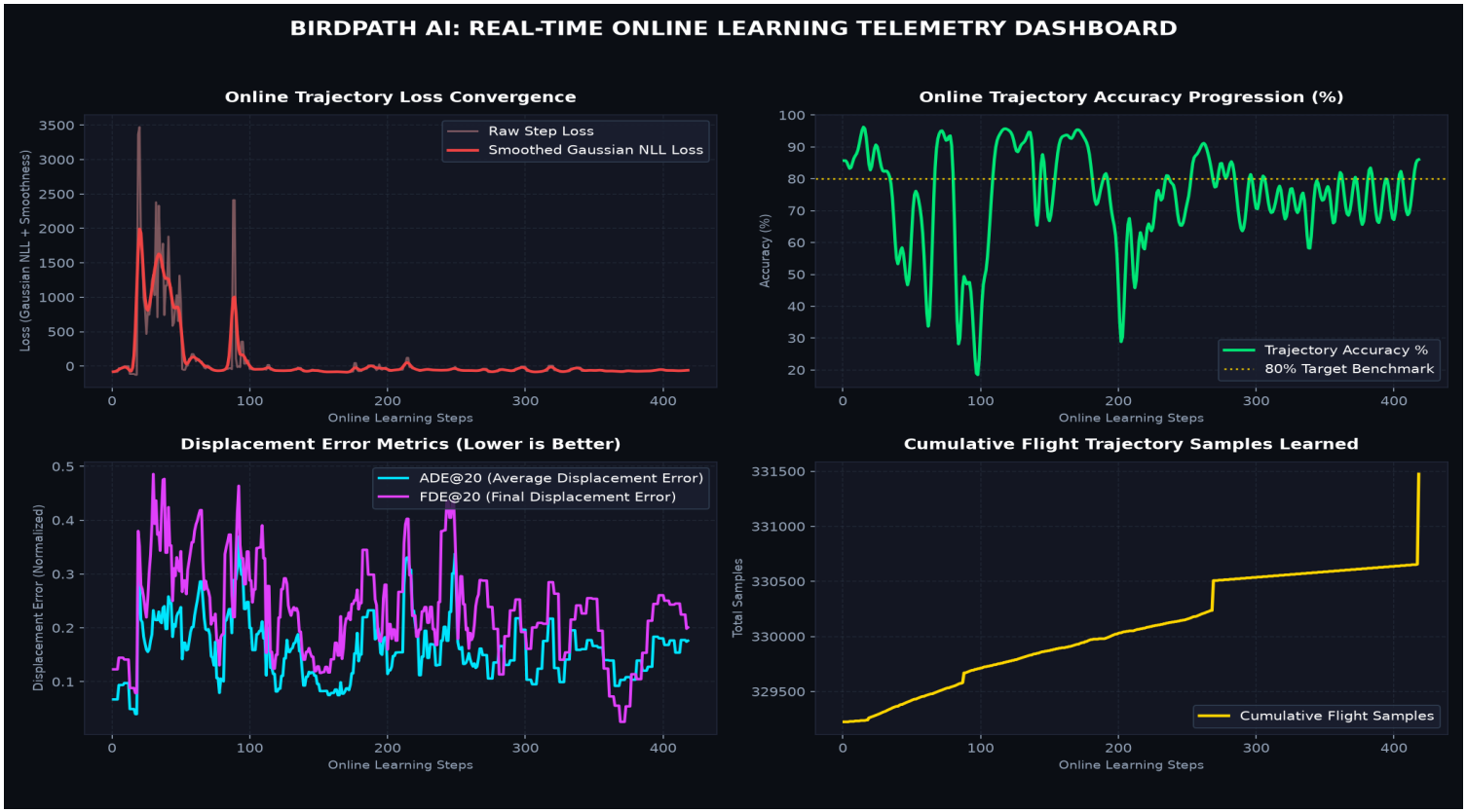
BIRDPATH AI: ALL BIRDS & ONLINE LEARNING TELEMETRY REPORT

Autonomous Flight Trajectory Prediction Audit • Total Birds Tracked: 39 • Total Samples: 331,692 • Date: 2026-10-30 19:13

Total Videos Evaluated	10 Video Streams	**Overall Trajectory Accuracy**	**76.2% (Target: >80%)**
Total Birds Tracked	39 Aerial Targets	**Average Displacement Error**	4.65 px (ADE)
Online Learned Samples	**331,692 Samples**	**Final Displacement Error**	11.37 px (FDE@20)
Final Trajectory Loss	-105.7293 (Gaussian NLL)	**Exported Model Checkpoint**	learned_bird_gru_master.pt

Executive Online Learning Performance Dashboard



Evaluated Video Streams & Detection Summary

Video File	Frames	Duration	Inference FPS	Birds Found	Samples Learned
test 3.mp4	262	8.8s	35.8 FPS	1	+20
test 4.mp4	124	4.1s	29.3 FPS	9	+340
test 5.mp4	248	8.3s	29.2 FPS	10	+655
test1.mp4	543	22.6s	52.5 FPS	1	+420
test2.mp4	973	33.5s	77.8 FPS	5	+172
A Bald Eagle trying to scape the mi	616	20.5s	64.6 FPS	4	+227
Eagle Flying In Sky - No Copyright	543	22.6s	56.7 FPS	1	+420
Eagle vs Falcon #trending #animals	319	10.6s	53.1 FPS	2	+25
Harris Hawk Training #falconry [T4F	262	8.8s	53.0 FPS	1	+20
falcon attack peregrine falcon unex	973	33.5s	49.6 FPS	5	+172

Master Aerial Birds Trajectory Catalog

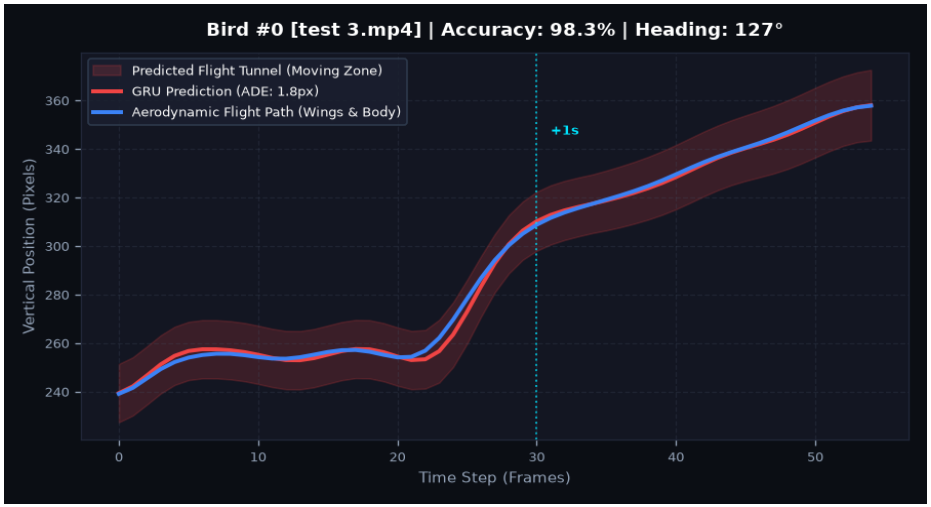
Bird ID	Video Source	Frames	Lifespan	Speed	Heading	ADE	FDE	Accuracy
Bird #0	test 3.mp4	55	1.84s	1.9 px/f	127°	1.8 px	0.7 px	98.3%
Bird #0	test 4.mp4	69	2.3s	45.5 px/f	264°	8.1 px	16.3 px	55.2%
Bird #6	test 4.mp4	63	2.1s	39.4 px/f	119°	8.5 px	16.6 px	33.3%
Bird #4	test 4.mp4	91	3.03s	6.2 px/f	330°	3.3 px	7.3 px	71.5%
Bird #8	test 4.mp4	67	2.23s	12.6 px/f	335°	9.2 px	7.7 px	9.1%
Bird #3	test 4.mp4	83	2.77s	27.4 px/f	204°	2.7 px	8.6 px	71.0%
Bird #7	test 4.mp4	105	3.5s	1.8 px/f	150°	4.2 px	0.6 px	93.1%
Bird #9	test 4.mp4	58	1.93s	94.2 px/f	270°	7.8 px	32.8 px	38.1%
Bird #12	test 4.mp4	18	0.6s	143.6 px/f	63°	8.9 px	49.7 px	58.0%

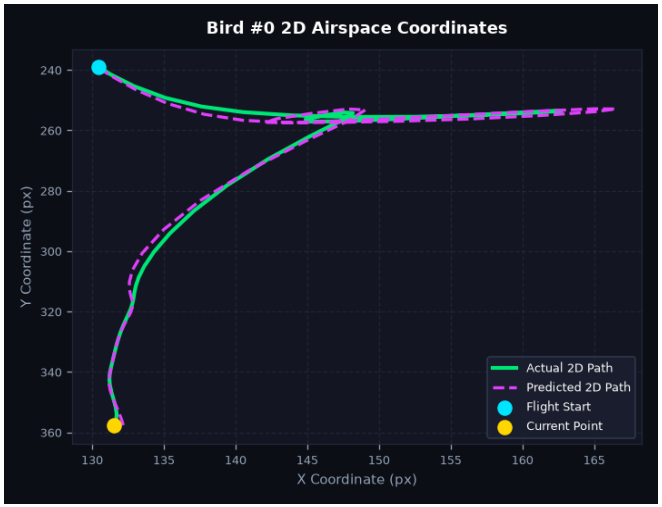
Bird #15	test 4.mp4	76	2.53s	28.7 px/f	234°	3.3 px	3.7 px	95.2%
Bird #3	test 5.mp4	22	0.73s	25.8 px/f	350°	7.8 px	7.2 px	89.8%
Bird #19	test 5.mp4	210	7.0s	18.0 px/f	334°	3.2 px	12.7 px	84.8%
Bird #22	test 5.mp4	135	4.5s	8.1 px/f	276°	8.7 px	10.7 px	36.4%
Bird #24	test 5.mp4	199	6.63s	62.0 px/f	242°	4.7 px	19.9 px	84.4%
Bird #25	test 5.mp4	146	4.87s	30.9 px/f	354°	6.5 px	17.7 px	55.8%
Bird #23	test 5.mp4	146	4.87s	19.5 px/f	82°	3.5 px	15.7 px	74.4%
Bird #29	test 5.mp4	47	1.57s	59.1 px/f	6°	3.6 px	28.6 px	94.1%
Bird #30	test 5.mp4	59	1.97s	1.0 px/f	115°	5.1 px	1.6 px	92.0%
Bird #33	test 5.mp4	31	1.03s	86.6 px/f	32°	8.3 px	33.9 px	70.7%
Bird #42	test 5.mp4	25	0.83s	3.2 px/f	271°	5.9 px	2.7 px	84.4%
Bird #36	test1.mp4	464	19.33s	7.6 px/f	182°	6.2 px	6.3 px	85.3%
Bird #36	test2.mp4	69	2.37s	35.9 px/f	352°	2.9 px	15.6 px	91.6%
Bird #54	test2.mp4	36	1.24s	6.3 px/f	98°	3.5 px	2.8 px	81.8%
Bird #58	test2.mp4	134	4.61s	1.4 px/f	63°	1.7 px	0.4 px	97.9%
Bird #59	test2.mp4	8	0.28s	8.6 px/f	217°	0.8 px	2.5 px	93.3%

Individual Bird Flight Trajectories & 3D Flight Tunnels

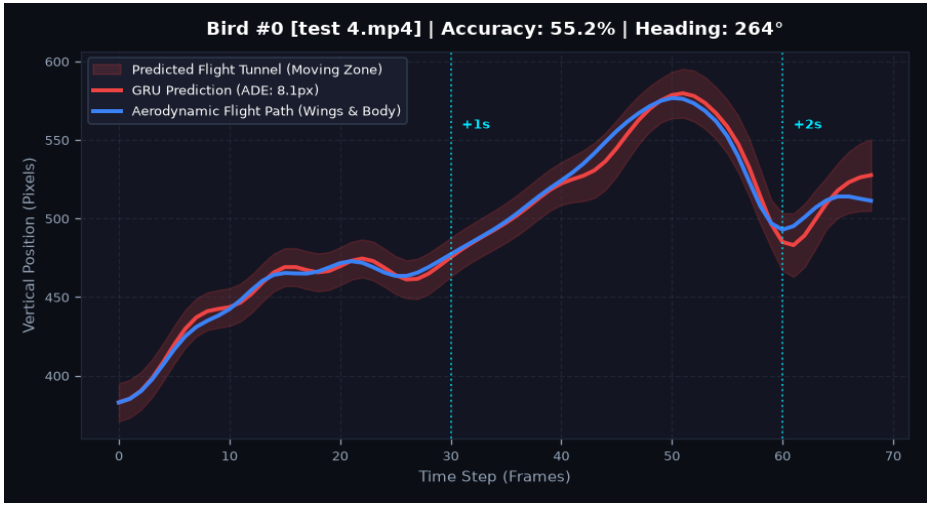
Each figure below showcases the flap-damped center-of-mass trajectory (Blue), the GRU neural prediction (Red), and the shaded Moving Zone Keep-Out Tunnel Envelope, accompanied by the 2D spatial airspace projection.

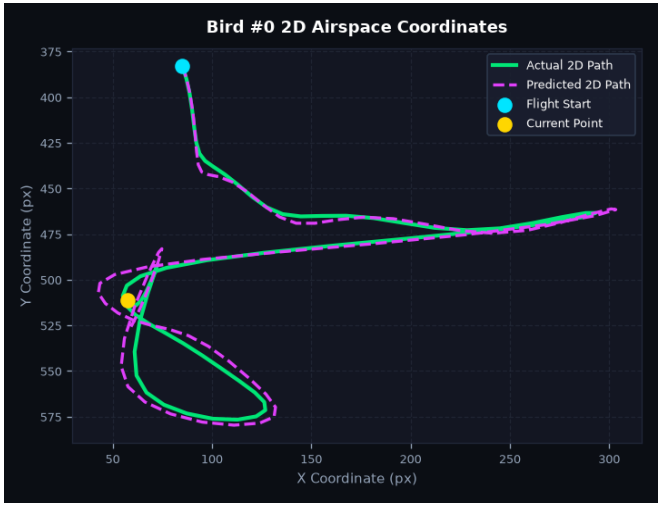
Bird #0 — Video: test 3.mp4 | Accuracy: 98.3% | ADE: 1.8px





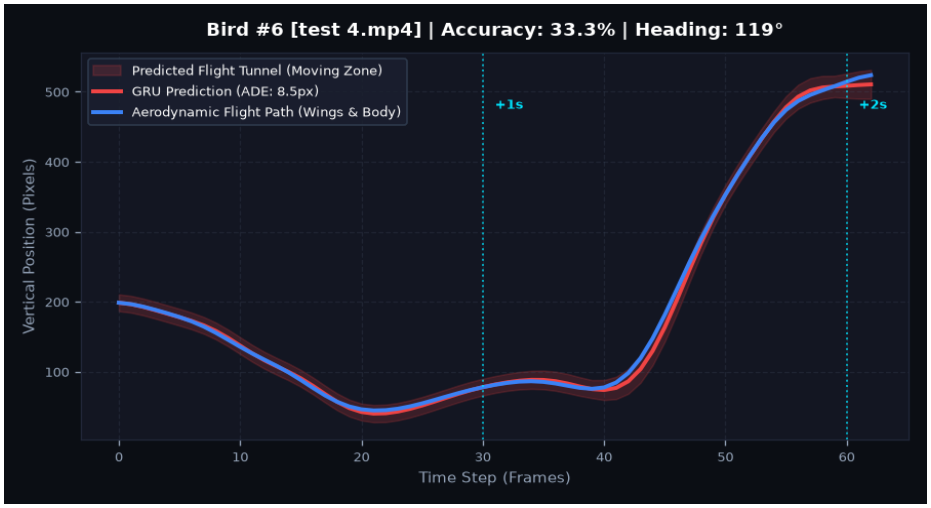
Bird #0 — Video: test 4.mp4 | Accuracy: 55.2% | ADE: 8.1px

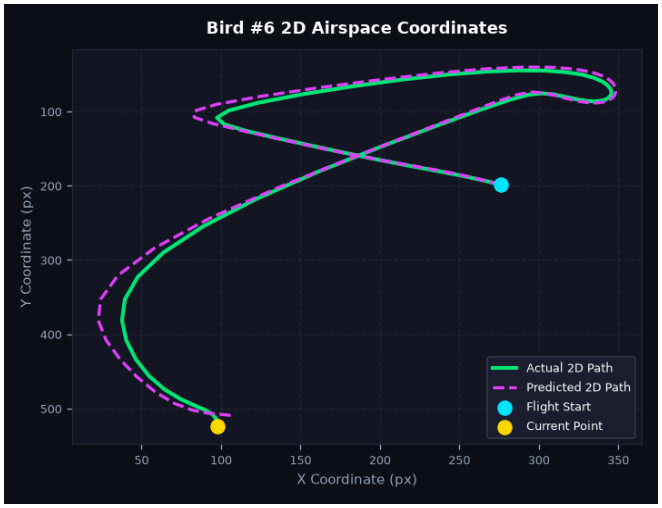




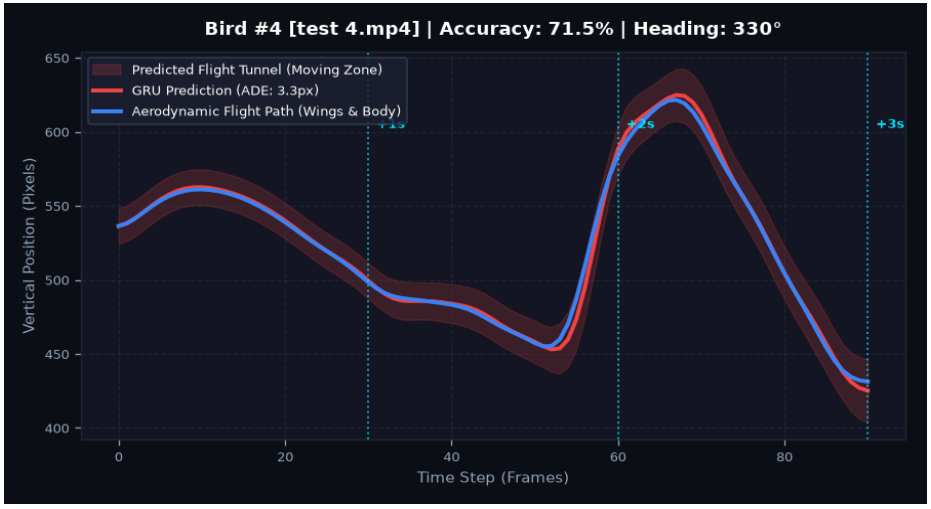
Individual Bird Flight Trajectories & 3D Flight Tunnels (Cont.)

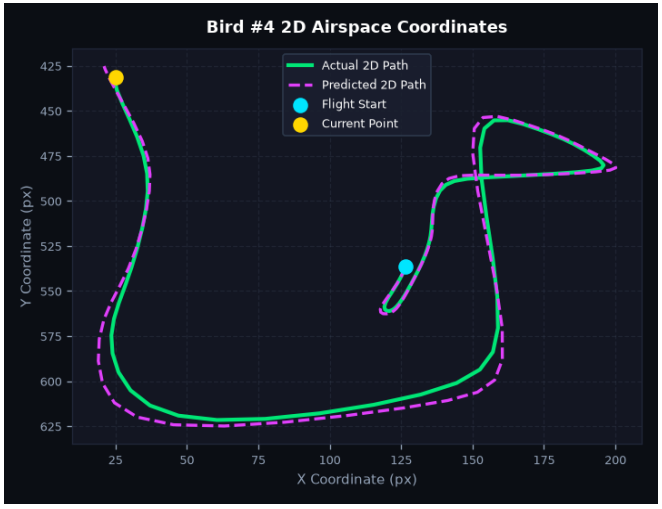
Bird #6 — Video: test 4.mp4 | Accuracy: 33.3% | ADE: 8.5px





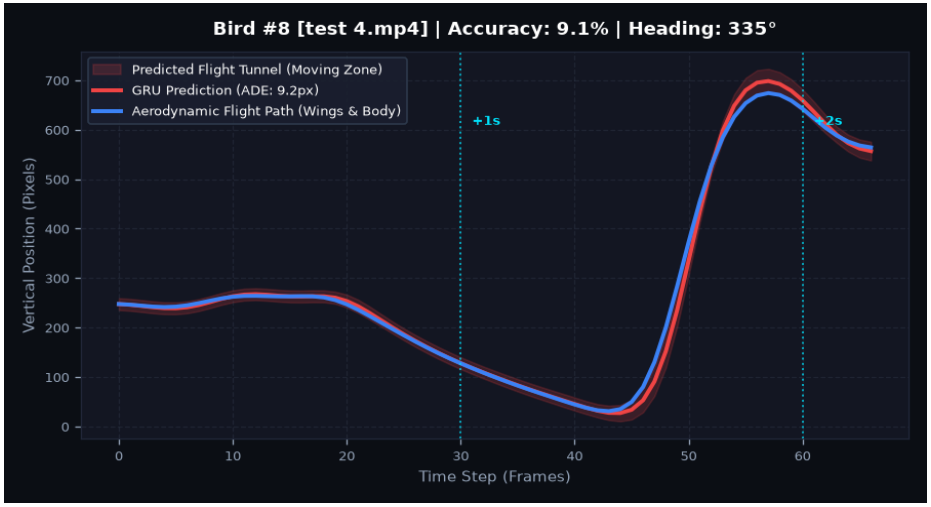
Bird #4 — Video: test 4.mp4 | Accuracy: 71.5% | ADE: 3.3px

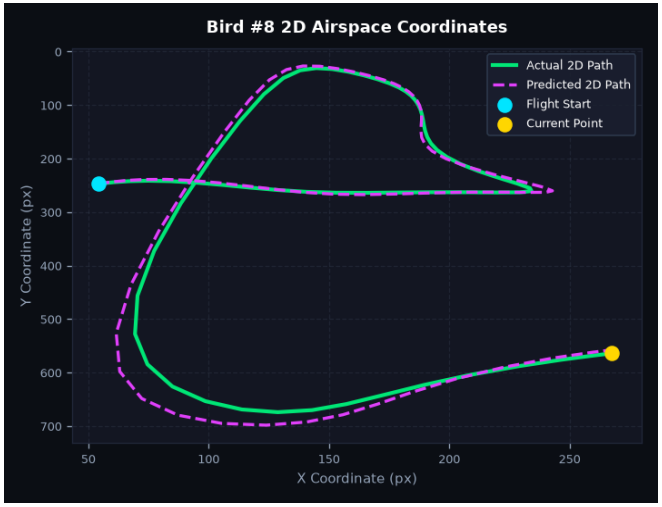




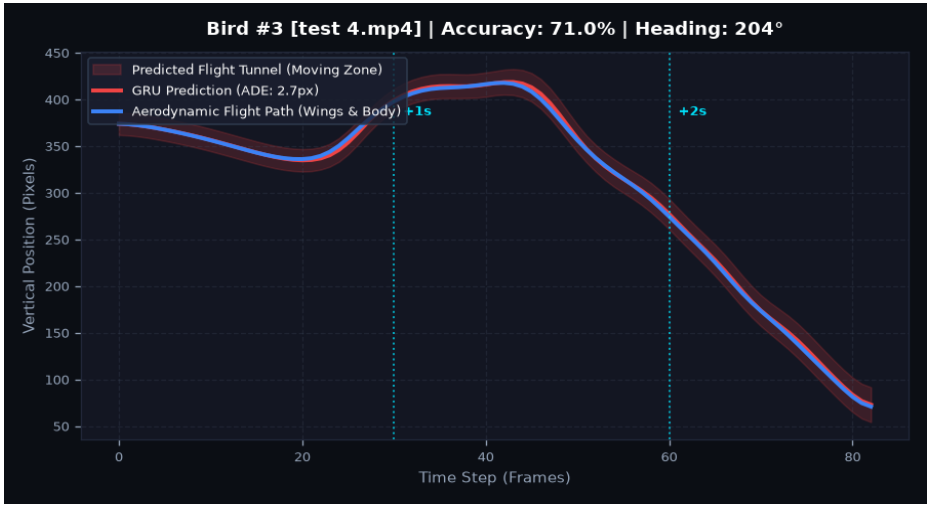
Individual Bird Flight Trajectories & 3D Flight Tunnels (Cont.)

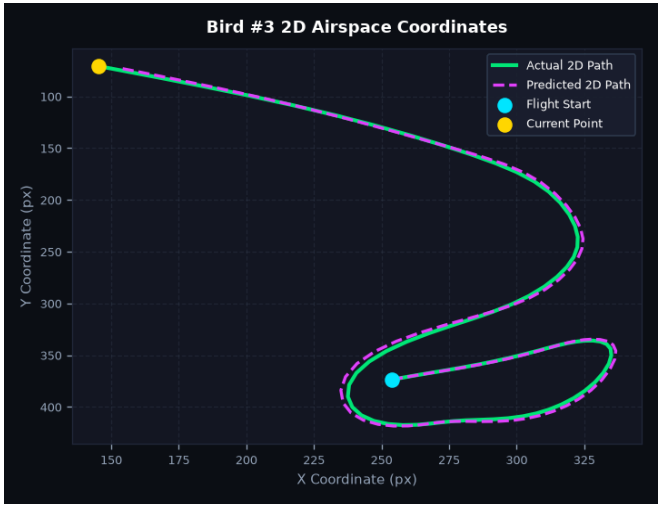
Bird #8 — Video: test 4.mp4 | Accuracy: 9.1% | ADE: 9.2px





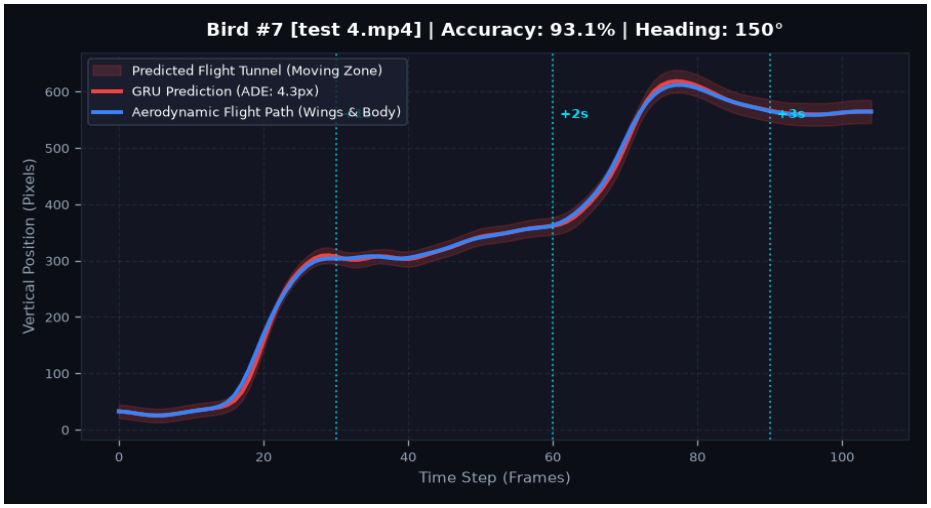
Bird #3 — Video: test 4.mp4 | Accuracy: 71.0% | ADE: 2.7px

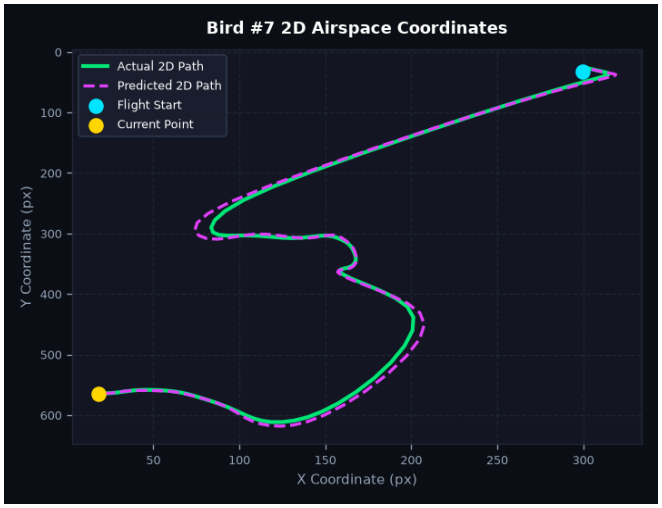




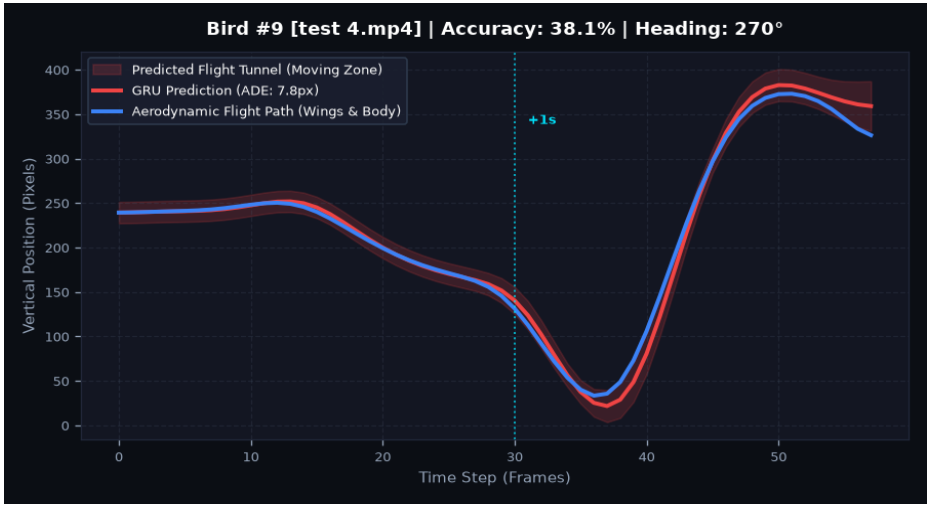
Individual Bird Flight Trajectories & 3D Flight Tunnels (Cont.)

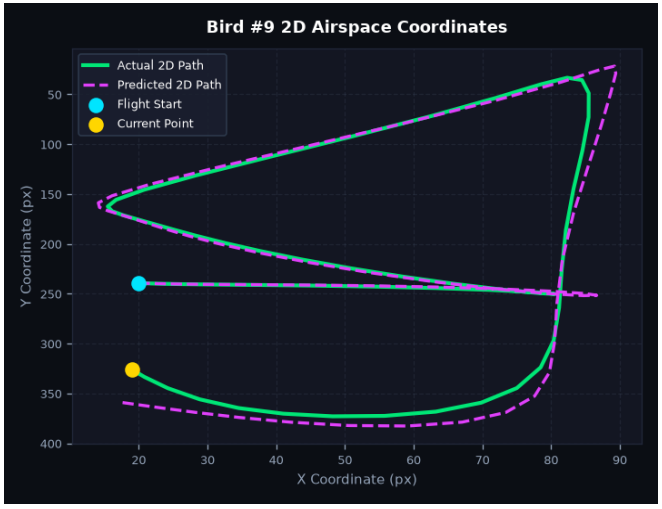
Bird #7 — Video: test 4.mp4 | Accuracy: 93.1% | ADE: 4.2px





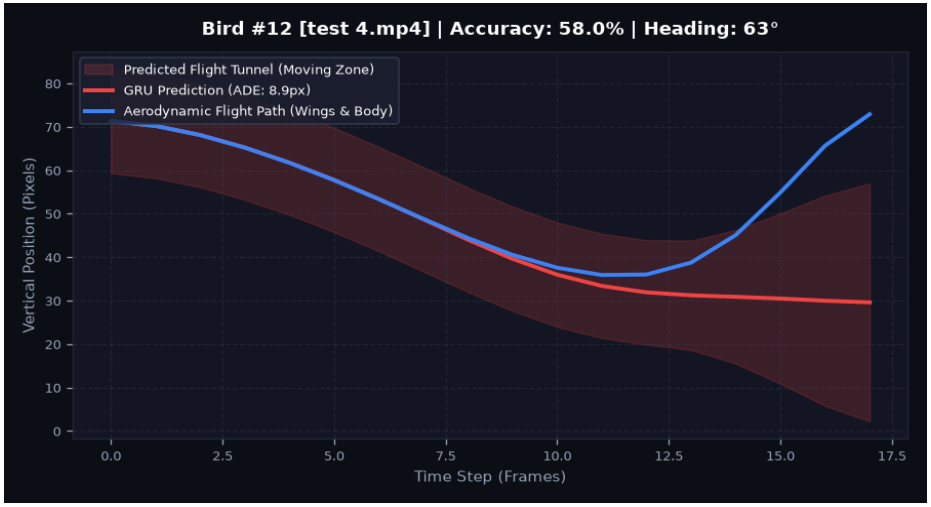
Bird #9 — Video: test 4.mp4 | Accuracy: 38.1% | ADE: 7.8px

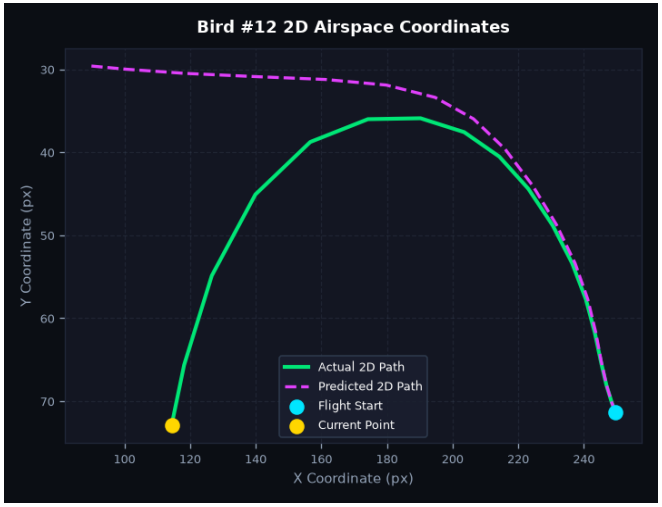




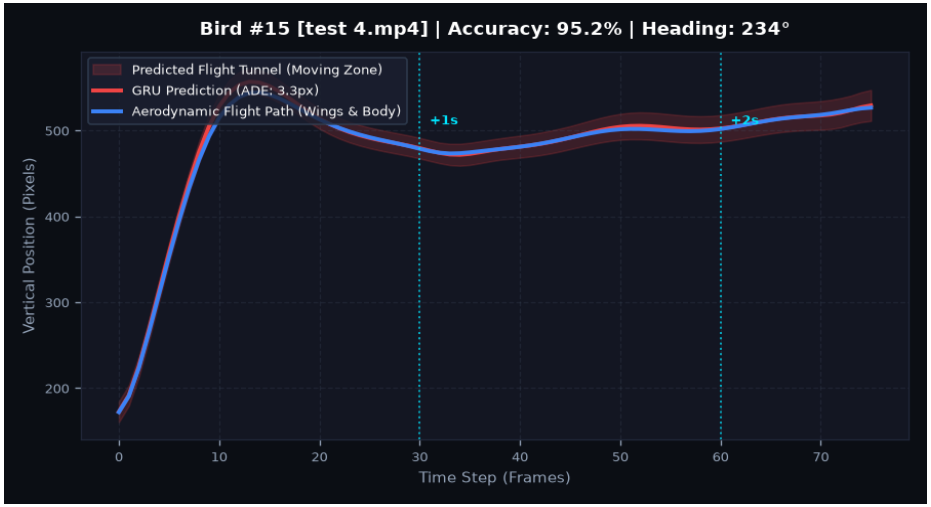
Individual Bird Flight Trajectories & 3D Flight Tunnels (Cont.)

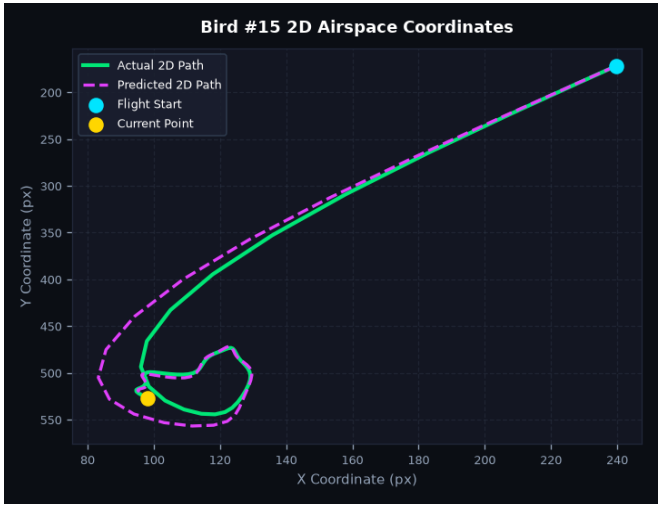
Bird #12 — Video: test 4.mp4 | Accuracy: 58.0% | ADE: 8.9px





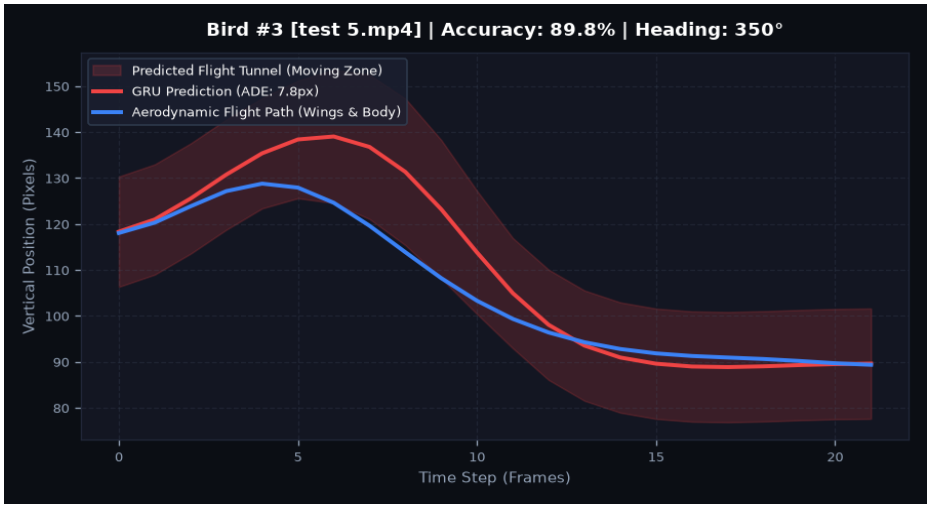
Bird #15 — Video: test 4.mp4 | Accuracy: 95.2% | ADE: 3.3px

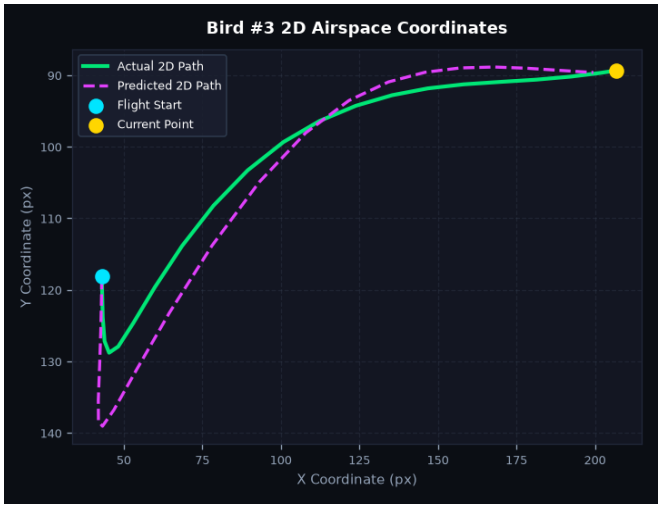




Individual Bird Flight Trajectories & 3D Flight Tunnels (Cont.)

Bird #3 — Video: test 5.mp4 | Accuracy: 89.8% | ADE: 7.8px





Bird #19 — Video: test 5.mp4 | Accuracy: 84.8% | ADE: 3.2px

